

3D ACADEMY

ACT SUBJECT TEST

QUESTIONS BANK

MATH LEVEL 1

Prepared by:

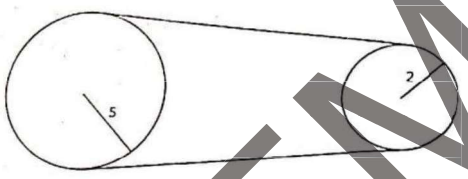
3D Academy Team

Test #1

1) The perimeter of a square is equal to the circumference of a circle whose area is 20 in^2. What is the area in square inches if the square?

A. 5π
B. 5
C. 20π
D. 20

2) The large circle has a radius of 5 ft and the small circle has a radius of 2 ft. The distance between the two centers of the circles is 9 ft. if the large circle, rotates 10 times the small circle rotates how many times?



A. 4
B. 10
C. 25
D. 50

3) A worker wants to make 10 pounds of mixed nuts using peanuts and cashews. He buys Peanuts at 3.75 dollars per pounds and cashews at 8 dollars per pound. If 1 pound of his mixture is valued at 5.45 dollars how many of peanuts did, he buy?

A. 6
B. 10
C. 7.25
D. 8

4) A garden with measurements 4m x 6m is surrounded by a sidewalk on all sides. The area of the garden with the sidewalk is 40 m^2 what is the width of the sidewalk?

A. $\frac{-5+\sqrt{41}}{2}$
B. $\frac{-5-\sqrt{41}}{2}$
C. 5
D. 41

5) Sarah invested 5000 dollars with 3% simple annual interest rate. What will be the interest in sarah's account after t years?

A. 15T
B. 150T
C. 3T
D. 1.5T

3D ACADEMY

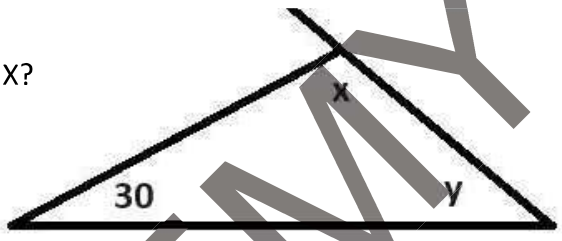
Test #1

6) The table below shows the number of bacteria increasing exponentially. What will the value of y be when x=10 ?

X	Y
0	1
1	2
2	4
3	8

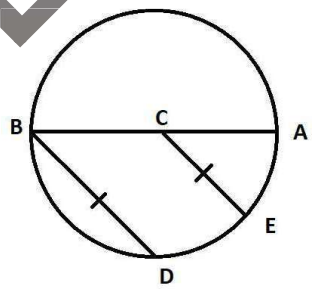
A. 124
B. 1024
C. 2015
D. 200

7) If X=2Y, which of the following is the value of X?



A. 60
B. 120
C. 30
D. 45
E. 100

8) IF Measure of Arc AED = 84, What is the Measure of Angle ACE?



A. 24
B. 84
C. 42
D. 168

9) Jade bought two similar planters. The second planter has dimensions 3 times as large as the first planter. if the first planter gets filled using 1 bag of soil, how many bags will the second planter need to be filled?

A. 9
B. 81
C. 27
D. 3

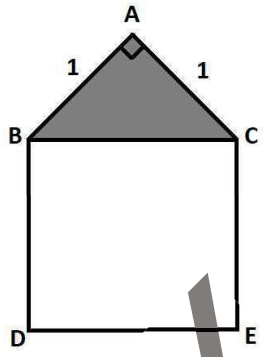
10) If $f(x) = y$, a point with coordinates (x,y) lies if the first number of the coordinates lies 3 less than half the second number?

A. $y = \frac{1}{2}x - 3$
B. $y = 2x - 3$
C. $y = \frac{1}{2}x + 3$
D. $y = 2x - 3$

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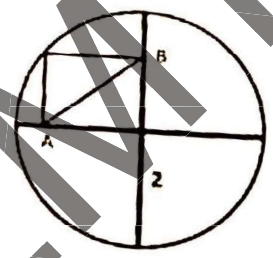
Test #1

11) Triangle ABC is isosceles. Find the Perimeter of ABCED.



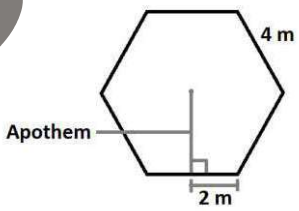
A. $2 + \sqrt{3}$
B. $2 + 3\sqrt{2}$
C. $2\sqrt{3}$
D. $3\sqrt{2}$

12) What proves $AB = 2$?



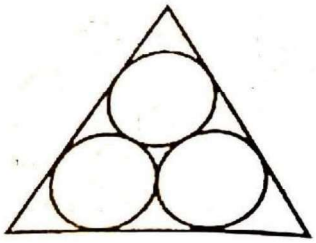
A. Diagonal of the rectangle are congruent
B. The radius = 2
C. The second diagonal = the radius
D. All the above

13) Find the area of a regular hexagon with side length 4 m. Round to the nearest tenth.



A. 83.1 m^2
B. 24 m^2
C. 41.6 m^2
D. 20.8 m^2

14) In this figure each of the 3 circles is tangent to the other. Each side of the triangle is tangent to 2 of the circles. Given that each circle has a radius of 3, what is the perimeter of the triangle?



A. $18 + 18\sqrt{2}$
B. $18 + 18\sqrt{3}$
C. $36 + 18\sqrt{2}$
D. $36 + 36\sqrt{2}$

15) Ali invested 5000$ in an account that paid him 32% compound interest, what will the balance of his account be after t years

A. $1600t$
B. $5000 + 1600t$
C. $5000(132)t$
D. $5000(1.32)^t$

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Test #1

16) What is the value of y for the solution to this system of equations $\begin{cases} 2x - y = 12 \\ y = x + 4 \end{cases}$

A. 8
B. 16
C. 20
D. 24

17) Simplify the following expression $-6x - (8 \div 2) - (4x - 5)$

A. $-10x + 1$
B. $-10x + 9$
C. $-2x9$
D. $-2x + 1$

18) If x and y are integers, which equation is always true

A. $\frac{x}{y} = \frac{y}{x}$
B. $x - y = y - x$
C. $x + 2y = y + 2x$
D. $2x + 3y = 3y + 2x$

19) In the geometric sequence (16, n, 36, …), The second term n is the geometric mean of 16 and 36. What is the value of n

A. 24
B. 25
C. 26
D. 28

20) John has x dollars. Ann has $5 more than john. If Ann gives john $10, then, in terms of x, how many dollars will Ann have

A. $x - 15$
B. $x - 10$
C. $x - 5$
D. $x + 5$

21) Automobile sales is United States dropped from 10.7 million unit one year to 9.3 million the next year. The decrease was closest to what percent

A. 1%
B. 13%
C. 14%
D. 15%

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Test #1

22) If the average of 5 consecutive whole numbers is 12, what is the largest number
   A. 14
   B. 16
   C. 17
   D. 22

23) solve the following inequality. If $5|8-2x|-3>27$
   A. $x<1 \; or \; x>7$
   B. $x\leq 1 \; or \; x\geq 7$
   C. $1<x<7$
   D. $1\leq x\leq 7$

24) Which of the following properties illustrate $3^x \cdot \frac{1}{3^x}=1$
   A. Commutative Property of Multiplication
   B. Associative property of Multiplication
   C. Property Multiplicative Inverse
   D. Identity Property of Multiplication

25) If x is greater than y and y is greater than 1, which of the following is LEAST
   A. $\frac{1}{x}$
   B. $\frac{1}{y}$
   C. $\frac{1}{x-y}$
   D. $\frac{1}{y-x}$

26) A store regularly sells books at 20% off the list price. At a sale its regular prices are reduced 10 percent. The sale price is what percent of the list price
   A. 30%
   B. 70%
   C. 72%
   D. 79%

27) The price of a cold roast beef lunch is $4.80, and the price of a hot roast beef lunch is $5.40 if both prices include 20 percent for tax and tip, what us the difference in price before tax and tip
   A. $0.12
   B. $0.40
   C. $0.50
   D. $0.60

3D ACADEMY

Test #1

28) For what price is 20 percent off the same $20 off

A. $1
B. $10
C. $100
D. $1,000

29) To which of the following is 6.06 closest

A. 6.1
B. 7
C. 6.7
D. 6

30) All of the following are greater than $\frac{1}{2}$ EXCEPT

A. $\frac{101}{200}$
B. $\frac{17}{33}$
C. $\frac{7}{12}$
D. $\frac{60}{1000}$

31) If $\frac{1}{4}$ of number is greater than 8, the number must be

A. Less than 2
B. Equal to 2
C. Less than 32
D. Greater than 32

32) If Ivan traveled 510 miles in 2 hours, 18 minutes, and 30 seconds. What was Ivan average speed

A. 221 mile/hour
B. 281 mile/hour
C. 229 mile/hour
D. 310 mile/hour

33) Which of the following numbers can be written in the form $(4 \times N) + 1$ where N is a whole number?

A. 55
B. 62
C. 71
D. 93

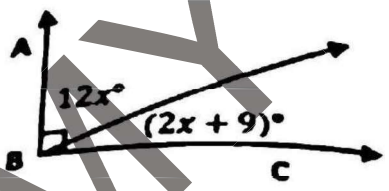
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Test #1

34) $if\ t = -3, then\ 3t^2 + 5t + 6$ equals

A. -36
B. -6
C. 6
D. 18

35) Based on the information in the figure, which statement shows a valid argument



A. (2x+9) + 12x = 180 because the angles from a linear pair
B. (2x+9) + 12x = 90 because the angles are complementary
C. (2x + 9) = 12x because the angles are vertical angels
D. (2x+9) = 12x because the angles are supplementary

36) According to a market study, City X has 1.75 millionaires for evert 1,000 people. if there are 8 million people in City, how many millionaires are there

A. 14
B. 140
C. 1,400
D. 14,000

37) What is the sum of the interior angels of a Hexagon?

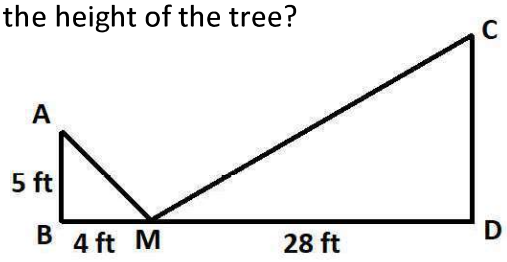
A. $720°$
B. 900°
C. 1080°
D. 1260°

38) For the line x + y = 1, what is the sum of the values of the slop, x-intercept, and y-intercept

A. -1
B. 0
C. 1
D. 2

39) Justine and Eric want to know the height of a tree. Eric places a mirror (M) on the ground in line with the top of the tree. He moves towards Justine represented by AB until she can see the top of the tree at point C. Justine's eye is 5ft from the ground at point A. To the nearest tenth of a foot What is the height of the tree?

A. 30
B. 31
C. 32
D. 35

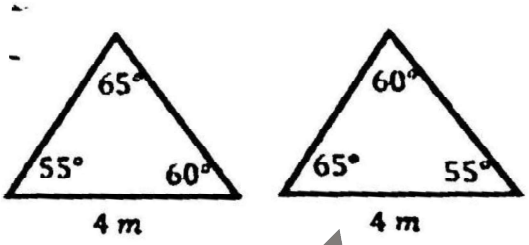


3D ACADEMY

Test #1

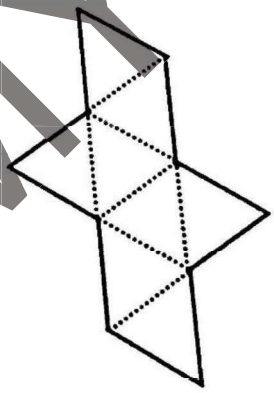
40) Which statement about these triangles is true?

A. The triangles are similar but not congruent
B. The triangles are not similar but not congruent
C. The triangles are similar and are congruent
D. The triangles are neither similar nor congruent



41) The net below represents a regular polyhedron or platonic solid how many edges does the platonic solid have?

A. 6
B. 8
C. 10
D. 12



42) If there are 9 white cards and 7 black cards, what is the probability of randomly picking a white card?

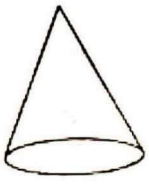
A. $\frac{9}{16}$
B. $\frac{9}{7}$
C. $\frac{7}{9}$
D. $\frac{16}{9}$

43) A circle has a diameter of 11 inches. What is the circumference of that circle?

A. π
B. $11\,\pi$
C. $5.5\,\pi$
D. $121\,\pi$

44) What is the cross section of this cone?

A. Circle
B. Oval
C. Rectangle
D. Triangle



3D ACADEMY

Test #1

45) $2(2x^3y^5z)^3 =$

A. $16x^9y^{15}z^3$
B. $16x^3y^{15}z^3$
C. $16x^9y^5z^3$
D. $16x^9y^{15}z$

46) $-2pq^3(\frac{1}{3}p^2q^5)^2 =$

A. $-2p^5q^{13}$
B. $-\frac{2}{9}p^5q^{13}$
C. $2p^5q^{13}$
D. $\frac{2}{9}p^5q^{13}$

47) Find the area of a circle with a diameter of 17.

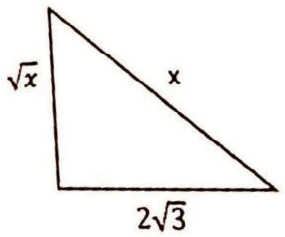
A. 227
B. 64
C. 22.7
D. 22

48) Factor $25x^2 - 121$

A. $(5x - 11)(5x - 11)$
B. $(5x - 11)(5x + 11)$
C. $(5x + 11)(5x + 11)$
D. $(x - 11)(x + 11)$

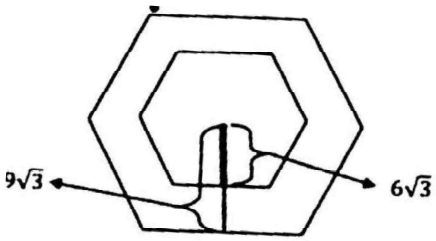
49) Find the value of x

A. 2
B. 3
C. 4
D. 6



50) Find the perimeter ratio of the large hexagon to the small one

A. $\frac{3}{2}$
B. $\frac{2}{3}$
C. $\frac{1}{2}$
D. $\frac{1}{3}$



3D ACADEMY

Test #1

Answer Keys:

Question	Answer	Question	Answer
1	A	26	C
2	C	27	C
3	A	28	C
4	A	29	A
5	B	30	D
6	B	31	D
7	E	32	A
8	C	33	D
9	C	34	D
10	A	35	B
11	B	36	D
12	D	37	A
13	C	38	C
14	B	39	D
15	D	40	A
16	C	41	D
17	A	42	A
18	D	43	B
19	A	44	A
20	C	45	A
21	B	46	B
22	A	47	A
23	A	48	B
24	C	49	C
25	D	50	B

3D ACADEMY